# NEXO-SMART EXPENSE SPLITTER


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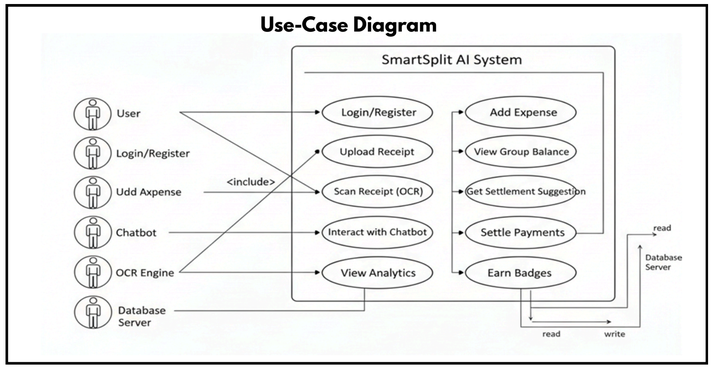
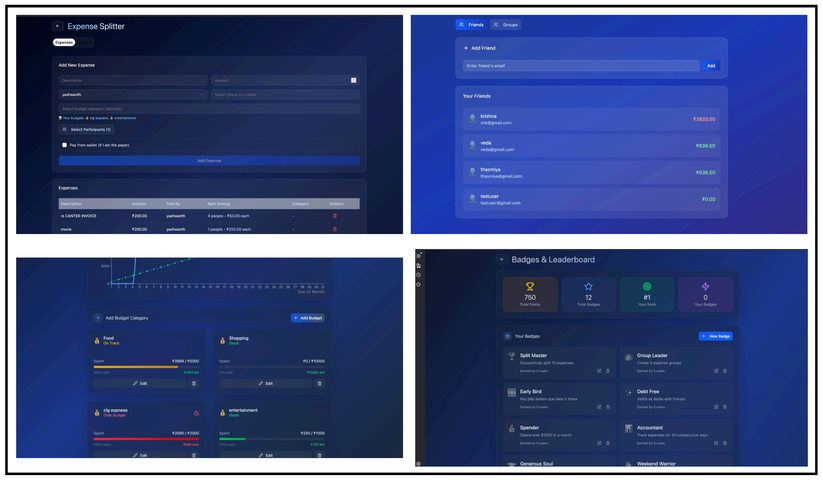
An intelligent and automated platform for managing shared expenses.

## SALIENT FEATURES:

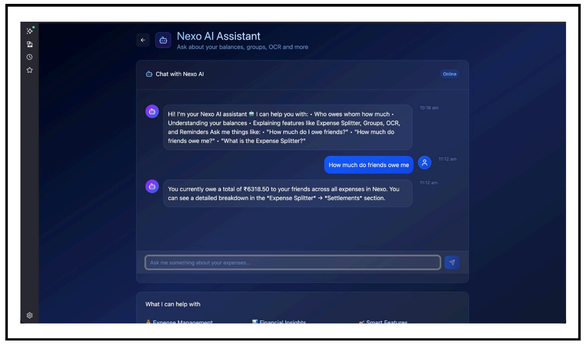
- Settlement Minimization: Implements a Greedy Graph Algorithm to calculate the absolute fewest number of payments required to clear all group debt, maximizing efficiency.
- AI/OCR Integration Readiness: Designed with a structured API to easily accept and utilize categorized expense data (Food, Travel, etc.) generated by external OCR/ML services.
- Secure Authentication: Utilizes industry-standard JWT (JSON Web Tokens) for protected, stateless user sessions and bcrypt for irreversible password hashing.
- Accurate Net Balance: Provides real-time, accurate balances by aggregating all financial movements (paid, owed, settled) into a single net debt/credit figure for every user.
- Group Collaboration: Offers essential APIs for creating groups, adding members, and recording settlements linked specifically to a group's financial history.
- An incorporated interactive AI Chatbot to answer user queries.
- Gamification Badges & Rewards are used to encourage good financial habits (timely payments, adherence to budgets) and enhance user retention.



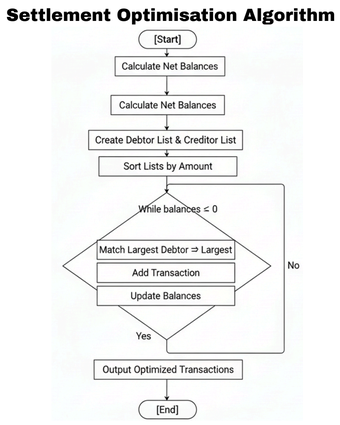
## FUTURE ENHANCEMENTS:

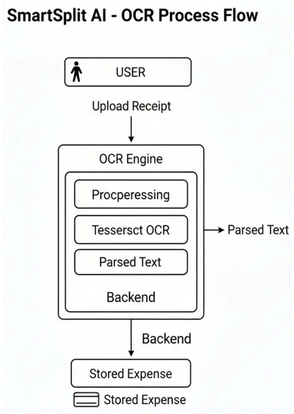
- Expanded Anomaly Detection: Additional AI modules could detect duplicate entries, irregular receipt patterns, suspicious transactions, or potential manipulation by members — enhancing trust and security.
- Mobile Application Build: Transition the responsive React/TypeScript front-end into dedicated mobile applications (iOS/Android) for optimal user experience and access.
- Payment Gateway Integration: Integrate third-party APIs (e.g., PayPal, Venmo, Stripe) to allow users to execute the optimized settlement payments directly from the application interface.
- The chatbot can be extended to support speech recognition and voice commands. Users could interact hands-free, such as by asking: "Hey SmartSplit, how much did I spend this month?"
- Multi-Currency and Localization Support
- Adding live WebSocket communication would allow instant synchronization of expenses and settlements without refreshing pages — useful during trips, events, or shared activities.
- Future upgrades could incorporate deep learning image classification to detect expense categories based on merchant type, receipt structure, and historical patterns.




## APPLICATIONS:

- Effortless Debt Tracking: Allows groups (roommates, travelers, families) to log shared expenses instantly, replacing complex spreadsheets and manual calculations.
- Guaranteed Accuracy: The transactional backend ensures that the final net balance calculated is 100% accurate, eliminating disputes over calculation errors.
- Secure Record Keeping: Provides a single, secure, cloud-based source of truth for all financial transactions and settlements within a group.
- Minimized Payments: Application of the Greedy Graph Algorithm to recommend the absolute minimum number of payments required, directly saving users time and bank transfer fees.
- Users are motivated to settle debts faster to earn the "Quick Payer" badge.
- Allow users to easily search for friends and add them to a group, streamlining the onboarding process for new collaborators.




## CONCLUSION:

The development of SmartSplit AI confirms the strategic feasibility of building a secure, intelligent expense system designed for operational efficiency. The project's most significant accomplishment is the validated establishment of a robust transactional core—moving the system beyond simple ledger management to guarantee data integrity through atomic SQLlite transactions. Crucially, the successful implementation of the Settlement Optimization Algorithm proves that computational intelligence can effectively solve real-world financial friction by minimizing debt transfer chains.